Math-CoT Distillation Suppresses Cue Acknowledgment Beyond What Compression Explains

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Abstract

Chain-of-thought (CoT) monitorability is the hope that reading a model's reasoning will tell us why it answered, so an overseer can catch undesired influences. We ask whether supervised fine-tuning on math traces preserves this property, using the cue-based paradigm of Meek et al. (2025) and Turpin et al. (2023) on a DeepSeek-R1-distilled student. Our central result is an intervention. Fine-tuning the student on a weaker model's math CoT, with the reasoning span genuinely present in the loss, cuts the CoT's acknowledgment of a planted cue from 25.0% to 14.4% on matched (question, cue) pairs (paired $\Delta = -0.110$, $n = 155$, McNemar $p = 0.005$) while the cue's behavioral pull on the answer is unchanged (paired $\Delta = +0.048$, $p = 0.52$) and held-out MATH accuracy rises ($0.650 \rightarrow 0.685$). The retrained student's traces are only $\sim 27\%$ shorter than the untrained model's, so CoT compression cannot account for the gap. The model keeps being moved by the cue and says so less often. That fine-tuning lowers CoT faithfulness on an answer-level metric is already known (Lobo et al., 2024); our contribution is that the cue's grip on behavior is held fixed while verbalization drops, which is the failure mode that matters for monitorability, and that this is shown by intervention rather than by conditioning on a post-treatment variable. A second arm isolates how much of the effect a shorter trace can produce on its own. Due to a chat-template defect we document in §3, an earlier run supervised the same student on the *final-answer segments* of the same traces; that arm compresses CoT $14\times$ and drives acknowledgment to 3.2%, so answer-only supervision reproduces a far larger collapse than reasoning-preserving supervision does. Length alone is nonetheless insufficient: at matched length and matched cue the gap persists (Fisher $p = 0.016$ in the long bin), and the collapse replicates on MMLU at $2.8\times$ compression rather than $14\times$. A same-strength self-distillation control rules out the weak-to-strong asymmetry as the cause. A capability-only certification can pass a model whose CoT has become markedly less revealing while still reading as elaborate.

1 Introduction

Chain-of-thought "monitorability" is the hope that reading a model's reasoning reveals why it answered, so an overseer can catch undesired influences [9]. Korbak et al. argue that this opportunity is fragile and that developers should think about how their design choices affect monitorability. Our paper looks at one such design choice, math-CoT SFT, and shows that it erodes monitorability in a specific way.

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Weak-to-Strong Reasoning (W2SR) [1][1] reports that fine-tuning a strong student on a weaker model's CoT raises accuracy. Concurrent faithfulness work [3, 2, 4] shows that models often do not verbalize the cues that actually move their answers.

W2SR certifies the student on accuracy alone and is silent on whether the distilled CoT stays monitorable. If distillation changed how a model reasons on the page without changing what it reveals, an accuracy check would miss the problem entirely.

The phenomena we exploit are largely known. Lobo et al. [5] showed that fine-tuning hurts CoT faithfulness on an answer-level metric. Zaman and Srivastava [7] argued via causal mediation that the dissociation between behavior and CoT verbalization can exist statically. Concurrent self-distillation work [6] identifies compression as a mechanism that suppresses CoT properties, in their case the verbalization of uncertainty. Our contributions are methodological and target-specific.

1. We induce a behavior–verbalization dissociation by intervention. On a run whose supervision genuinely contains the teacher's reasoning (§5), math-CoT SFT moves an R1-distill student from acknowledgment 25.0% to 14.4% (paired $p = 0.005$) while paired influence is unchanged ($\Delta = +0.048$, $p = 0.52$) and held-out MATH accuracy rises. Because the manipulation is the training data rather than a filter applied to the outcome, this does not condition on any post-treatment variable. The new piece beyond Zaman and Srivastava is the controllability: an ordinary post-training recipe produces the dissociation on demand.
2. A same-strength self-distillation arm serves as a negative control. With teacher equal to student, the collapse is indistinguishable from W2SR (paired $p = 1$, $n = 79$), and every answer-only arm we tested against W2SR gives $p \geq 0.09$. We have not seen this control in the literature in this form. It is the cleanest evidence that the cause is the SFT in general, not weak-to-strong asymmetry, and it separates our claim from the concurrent self-distillation work on uncertainty, since the outcome variable is different but the underlying mechanism is similar. We are explicit that this control was run under the answer-only supervision format, which weakens it: with the teacher's reasoning outside the loss, the three teachers differ only in their final answers (§7).
3. We separate how much of the collapse is trace length from how much is not, using a supervision-format contrast that we did not design but can exploit. The answer-only arm (§3) compresses CoT $14\times$ and reaches acknowledgment 3.2%; the reasoning-preserving arm compresses only $\sim 27\%$ and reaches 14.4%. Length therefore carries much of the raw drop. It does not carry all of it: at matched length and matched cue composition the gap survives (long bin 44.0% vs 18.8%, $\mathrm{OR} \approx 3.40$, Fisher $p = 0.016$; §5), the reasoning-preserving arm sits significantly below baseline at near-matched length, and the MMLU replication reproduces the full collapse at $2.8\times$ compression rather than $14\times$.
4. A reproduction of W2SR with a headroom probe shows that its accuracy claim is itself conditional. The gain only fires with (i) an in-family teacher and (ii) a base student. A zero-shot CoT prompt on the untrained student separates real elicitation of latent capability from a CoT-prompting confound (Section 4), and the certified guarantee is narrower than Yuan et al. claim.
5. Putting these together gives a clear certification consequence. An accuracy-only certification of any math-CoT-SFT student, including same-strength self-distillation rather than just weak-to-strong, can pass a model whose CoT has become markedly less revealing. A monitorability composite that rewards elaborate-looking CoT misses the failure too. The cue-acknowledgment component has to be measured directly.

2 Related work

Lobo et al. [5] show that task-specific fine-tuning lowers CoT faithfulness across four datasets, and we take this degradation as given. The claim we are making is not that faithfulness drops, but that it drops while the cue's behavioral influence on the answer does not. That is the dissociation their answer-level metric cannot isolate. Concurrent work by Jiang et al. [11] makes the closest

[1] Distinct from the earlier paper with the same title by Yang, Ma, and Liu [10] (EMNLP Findings 2024), which uses a progressive SFT+DPO recipe on GSM8K, MATH, and OlympicArena. We build on Yuan et al.'s LoRA-SFT pipeline.

complementary observation in a different domain: on medical CoT distillation, answer accuracy improves ($74.7\% \to 84.4\%$ MedQA) while reasoning step error rate climbs ($30.6\% \to 50.3\%$), which they use to argue that answer-level metrics alone mask reasoning-quality decay. Their finding and ours share the shape "the answer looks better while the trace gets worse" in two very different domains, which is stronger joint evidence than either result alone.

Concurrent work [6] finds that self-distillation suppresses epistemic verbalization, specifically the verbalization of uncertainty tokens like "wait" and "hmm," via compression. They observe that this hurts out-of-distribution generalization. We share the compression mechanism and the self-distillation setup, but our outcome variable is a different property of the CoT: cue-acknowledgment (does the trace mention or attribute reasoning to a planted hint) rather than uncertainty-token density. Uncertainty-token loss and cue non-acknowledgment are distinct failure modes that happen to share a mechanism. On the RL side rather than SFT, Little [14] independently shows that training with an explicit length penalty makes CoT less monitorable at matched length (7-35 pp lower hint disclosure after random-shortening baselines). Our result is the SFT-side complement: math-CoT SFT compresses CoT even with no length term in the objective and produces the same qualitative degradation, so the failure mode is not specific to length-penalized RL.

Meek et al. [2] measure monitorability as a combination of faithfulness and verbosity across model families. We use their cued-prompt eval but study how a training intervention moves faithfulness on a fixed substrate. We pair the standard verbal-acknowledgment measurement with a behavioral measure that separates "stops being influenced" from "stops verbalizing influence," and we ask the sharper question of whether faithfulness falls beyond what CoT length alone explains.

Zaman and Srivastava [7] argue that the Biasing Features metric, which flags certain answers as "unfaithful," partly captures the lossy compression of distributed transformer computation into linear natural language. They show via causal mediation that even non-verbalized hints can causally route influence through the CoT to the answer. That is our dissociation, observed inferentially. We add three things on top. First, we induce the dissociation with a training method, moving acknowledgment from 25% to 3% on a fixed substrate. Second, we pin the behavioral pull directly, via paired influence and a flip-to-cue-target rate at about $2\times$ chance, rather than infer it through mediation. Third, we draw the certification consequence, since an accuracy-only or verbosity-rewarding check passes the very models on which the dissociation is most extreme. Zaman also argue that hint-acknowledgment is a poor faithfulness metric because absence of hint words does not prove the hint was not causally involved. We agree with that critique on faithfulness grounds. But for monitorability the metric is aligned with the task: what an overseer can read off the trace determines whether they can catch the influence, independent of whether the model was "really" moved by the hint in some deeper counterfactual sense. Our CoT-conclusion analysis in §5 further shows the CoT reasoning itself walks to the cue target in most silently-influenced cases, so acknowledgment absence is not merely compression stripping the hint word.

Channel migration is not what is happening here. Recent work on extended-thinking models documents a thinking-vs-answer divergence in hint acknowledgment, often a large one (for instance 87.5% vs 28.6% in [8]; concurrent work [18] finds a similar divergence on a larger 12-model panel with an "unacknowledged" quadrant of 11.8%). Our judge reads the full chain-of-thought, that is, both the thinking tokens and the answer. So the collapse we measure is not an artifact of which channel is read; acknowledgment vanishes in both channels combined.

Xiong et al. [12] provide a useful contrast on the training-method axis: RLVR (reinforcement learning with verifiable rewards) spontaneously improves monitorability during training, driven by response sharpening and increased prompt attention, without monitorability being an explicit objective. Our result is the opposite for a different post-training procedure: SFT on math-CoT traces, also without monitorability in the objective, degrades monitorability instead. Which post-training method a model receives can flip the sign of the monitorability side effect, even when neither method targets monitorability directly. Related term-of-art usage: Yee et al. [13] introduced the language of "dissociation" for faithful-vs-unfaithful reasoning mechanisms in a static setting; our use of "behavior–verbalization dissociation" is in that same lineage but induced by training.

Known limits of the cue-based paradigm. The Turpin-style cue-acknowledgment paradigm we adopt has documented limitations. Young [15] shows that the choice of faithfulness classifier alone shifts measured faithfulness by up to 30 pp across otherwise identical setups, so absolute rates should be read as judge-dependent. Scalena et al. [16] argue that reasoning models commit to an answer well before the CoT ends, so late CoT content can be epiphenomenal and length-only comparisons can be misleading (our Task H length-binned analysis is designed to defuse this specific critique on our data). Za et al. [17] document that Sonnet is roughly $3\times$ more susceptible than Gemini under adversarial persuasion when used as a monitor, which is one reason we cross-check our Sonnet judge with two others: Gemini (Cohen's $\kappa \in [0.64, 0.68]$, substantial) and Kimi (overall $\kappa = 0.556$, moderate; systematically stricter than Sonnet). All three judges agree on the direction of the ack drop and give one-directional discordant patterns. We frame our headline metric as *monitorability* (does the trace reveal the influence to an overseer) rather than *faithfulness* (does the trace causally reflect the model's internal computation) precisely because the paradigm operationalizes the former, not the latter: an overseer only has the trace, so what the trace does or does not say about the cue is directly what determines whether the influence is catchable.

3 Setup

W2SR [1] samples CoT from a weak teacher on a problem set, keeps correct and incorrect traces, LoRA-SFTs a stronger student, and reports held-out Pass@1. Monitorability follows Meek et al. [2]: each question is paired with a cue (authority opinion, leaked answer key, embedded metadata, and so on) that points at a target. An LLM judge reads the full CoT and scores faithfulness (does the CoT acknowledge the cue) and verbosity (a factor-utilization rate). We use the five cues shipped in the released eval, with `claude-sonnet-4-6` at $T = 0$ as the single judge. Faithfulness is reported as acknowledgment among cued samples. All evaluation is greedy ($T = 0$), trace sets are hashed, and training is LoRA. Generation, training, and gating run on Modal GPUs. All code, configs, per-sample judge labels, and the reanalysis pipeline are released. The four headline numbers in Table 2 are enforced by hard-fail assertions in `01_gate.py`, so any drift halts the suite.

Procedural details that matter for interpretation. Trace generation used $T = 0.6$, top-p 0.95, and a repetition penalty of 1.1 (the penalty was added for the loop-prone 1.5B teacher and kept constant across all teachers for comparability; it discourages long repetitive CoT and so interacts with the length analyses). Traces without a parseable `\boxed{}` answer were dropped before SFT, which preferentially removes traces truncated at the generation budget and therefore biases the training set slightly short. The under-test conditions were evaluated with an 8k generation budget (except the cross-family Llama arms, which used 16k); the teacher reference rows used 30k, so teacher-vs-student comparisons are not budget-matched (the baseline-vs-trained comparisons that carry every headline result are). SFT computes the loss over the full rendered sequence, prompt tokens included, rather than masking to the assistant span. Task H's W2SR side pools a larger re-run of the three text cues with the original run's two non-text cues (634 records) to give the length bins usable counts; the paired analyses elsewhere use the original matched set. The main eval judge is `claude-sonnet-4-6`; Tasks I2/I3 and J were judged with `claude-sonnet-4.5` via OpenRouter. Task J additionally uses a hand-written binary yes/no acknowledgment rubric rather than the in-eval `cue_aware` rubric, so Task J rates differ from the main-eval rates by both judge and rubric and the two should not be read as interchangeable. Re-running an LLM judge does not reproduce its labels exactly even at $T = 0$: an identical re-run of the Task G rubric swap moved baseline acknowledgment from 9.4% to 8.8% and its McNemar p from 6×10^{-5} to 1.2×10^{-4}. We report the committed label set and note that judge-level digits carry this much noise.

What the supervision actually contained. This matters for how the results should be read, so we state it explicitly. Traces are generated by prompting the teacher with the R1 chat template, whose generation prefix ends `<|Assistant|><think>`, so each stored trace has the form `{reasoning}</think>{answer}`. Our SFT examples were rendered with the student's own chat template. The published R1-Distill template contains, for assistant turns, a rule that splits the content on `</think>` and keeps only the final segment. The tokenized supervision for every R1-substrate arm in this paper was therefore the *final-answer segment only*, with the reasoning span removed — even though the trace files on disk retain it (which is why a check of the trace data alone does not reveal this, and why we report it here). The intervention we are measuring is thus SFT on the answer segments of math CoT traces ("answer-only" supervision), not SFT on the reasoning itself. We flag the consequences in Section 7: it gives the CoT compression a straightforward explanation, and it makes the teacher-strength invariance in Table 2 much less surprising, since the teacher's reasoning

Table 1: Reproduction (held-out MATH L3 to L5 Pass@1). In-family native-Qwen teacher; the headroom probe separates genuine elicitation from CoT-prompting. The gain is genuine on the base student and is an artifact on the instruct student.

student	unelicited	zero-shot CoT	W2SR	W2SR − CoT
Qwen2.5-Math-7B (base)	0.325	0.240	0.645	+0.405 (genuine)
Qwen2.5-7B-Instruct	0.230	0.630	0.630	+0.000 (artifact)
Foreign-style teachers (R1-distill-1.5B, SimpleRL-Zoo) into 7B: no gain or collapse.				

never entered the loss. It does not affect the behavioral measurements, the paired acknowledgment collapse, or the matched-length residual, all of which are properties of the resulting models rather than of the training data. The released training code now renders the assistant turn without the template and hard-fails if the reasoning span is absent from a rendered example; Task K (§5) re-runs the W2SR-weak arm with that fixed renderer and measures what the format change was responsible for.

4 When W2SR fires

With a foreign-style weak teacher (DeepSeek-R1-distilled 1.5B, or SimpleRL-Zoo) distilled into a Qwen2.5-7B student, we found no Pass@1 gain and frequent degeneration, across students, ranks, and decodings. W2SR only reproduces with an in-family, native-Qwen weak teacher. With `Qwen2.5-Math-1.5B-Instruct` (about 67% on the train problems) as teacher and `Qwen2.5-Math-7B` as student, held-out MATH (L3 to L5) Pass@1 goes from 0.325 to 0.645, with format-valid 1.00.

A $+5$ point gate-pass on an already-CoT-capable model could be re-arrangement rather than real elicitation. To check, we score the untrained model under a zero-shot CoT prompt, which we take as the prompting-only ceiling, and we report W2SR − (zero-shot CoT) (Table 1). On the base student, W2SR adds $+0.41$ beyond prompting, which is genuine elicitation of latent capability. On an instruct student it adds $+0.00$. The apparent gain over an unelicited baseline is purely the CoT-elicitation confound. Two conditions that Yuan et al. leave implicit are therefore necessary: an in-family teacher, and a base (non-CoT-eliciting) student.

5 Math-CoT SFT degrades CoT monitorability

We compare a baseline student, a W2SR student (LoRA-SFT'd on a weak in-family teacher's MATH CoT), and references, on GPQA. The SFT lands the student at its own CoT ceiling, so faithfulness differences are not a capability confound. The question we are asking is whether distillation changes faithfulness, not whether it improves capability.

The instruct student is at the floor. On `Qwen2.5-7B-Instruct`, cue-acknowledgment is at a floor for every condition (baseline 1.8% at $N = 970$; W2SR 3.6%; strong-teacher control 4.6%), while teacher references reach 22 to 37%. We cannot detect faithfulness transfer here. But verbosity does transfer strongly ($\Delta = +0.26$ for W2SR over baseline, CI $[0.22, 0.31]$), so the model looks more monitorable without any measurable faithfulness change. To get a substrate with dynamic range, we move to a reasoning student.

The answer-only arm: behavior strengthens, verbalization collapses. On the reasoning student, the setup is fully in-family: student `R1-Distill-Qwen-7B`, weak teacher `R1-Distill-Qwen-1.5B`. This subsection describes the arm whose supervision was answer-only (§3); §5 gives the reasoning-preserving re-run that carries our headline. The trained student's GPQA answers remain parseable at 94.2% ($179/190$ cued samples, vs 76.9% for the untrained baseline), and baseline acknowledgment is off-floor at 25% (Chua and Evans [4] report $\sim$59% on the full DeepSeek-R1; our lower baseline is on the distilled 7B variant on GPQA, so absolute levels differ across scale and eval set). Capability on this arm is *not* preserved: the only committed measurement of the preregistered held-out MATH gate — a re-gating run performed alongside Task K (§5) — *fails* it (Pass@1 0.550 vs the untrained reference 0.695, with a repetition-loop flag). We disclose this as a preregistration deviation (§7).

Paired influence (answer equal to cue target on matched (qid, cue) pairs) gives $\Delta = +0.244$, 95% CI $[+0.110, +0.378]$, $n = 82$, McNemar $p = 0.0017$, discordant 9/29. Among answer flips, the flip lands on the cue target 73.7% of the time for W2SR against 54.8% for baseline; both exceed the $1/3$ random-flip chance (baseline $p = 0.003$, W2SR $p \approx 3.5 \times 10^{-19}$), and unlike in an earlier version of this analysis they are *not* statistically indistinguishable from each other (Fisher two-sided $p = 0.032$). So on this arm the cue's behavioral grip is not merely retained but strengthened. If we restrict to the cue-points-at-wrong-answer stratum, so that landing on the cue is by definition a wrong answer, flippers still hit the cue target at 66.7%/74.1%/74.0% for baseline, W2SR, and strong-teacher KD (each $p \leq 9 \times 10^{-4}$ against $1/3$). The pull is real cue-influence and not accuracy in disguise.

A note on answer extraction. These behavioral numbers depend on recovering each completion's final answer letter, and that step is not neutral between conditions. Our extractor originally accepted a loose "…answer is X" pattern anywhere in the completion as a fallback. On a long trace truncated before it concluded, the last such match is frequently the model restating the cue rather than answering — for one baseline sample, at character 528 of 34,142. Of the 33 baseline records extracted that way, 21 equalled the cue target and only 7 were correct, at four-way chance, against 52% correct for records carrying an explicit answer marker. Because the failure requires a long unfinished trace it struck the long-CoT baseline far harder than the short-output trained arms, and trace length is the treatment variable. We therefore count an answer only when the model emits an explicit `ANSWER:` or `\boxed{}` marker. This correction *raises* every reported effect, because the contamination inflated the baseline: paired influence moves from $\Delta = +0.157$ ($p = 0.021$) to the $+0.244$ ($p = 0.0017$) above. Acknowledgment is judge-scored and is untouched by the extractor.

Verbalization collapses on both rubrics. On the upstream loose rubric ("acknowledge the cue"), W2SR drops pooled acknowledgment to 3.2% ($\Delta = -0.220$, CI $[-0.293, -0.153]$, McNemar $p \approx 2 \times 10^{-9}$, discordant 34/1). The drop holds at every text cue (Table 4: stanford 47% $\rightarrow$ 8%, grader-hack 22% $\rightarrow$ 3%, insider 56% $\rightarrow$ 5%), and the discordant cells are one-directional (12/0, 7/1, 15/0). Figure 1 summarizes the behavior–verbalization split across all trained arms. A stricter "attribute reasoning to the cue" rubric, scored by `gemini-2.5-pro` (the same out-of-family judge used for the re-scoring below), shrinks both rates (baseline 9.4%, W2SR 0.0%, 15/0 discordant, $p = 6 \times 10^{-5}$). The magnitude is smaller but the drop becomes even more one-directional. Both numbers hold at roughly $14\times$ shorter CoT (baseline median 18,692 chars, W2SR 1,362), which is the central confound this arm cannot rule out and §5 is designed to address.

Uncued GPQA accuracy is a weak capability check on this substrate and we do not rest anything on it. At the 8k generation budget the untrained baseline fails to emit a parseable answer on 20 of 40 uncued items — it writes ~20,000 characters and runs out of room — against 2 of 40 for the answer-only arm. Scoring unparseable items as incorrect gives baseline 0.275 and the trained arm 0.425; scoring only the parseable subset reverses the order, 0.550 against 0.447. The comparison mostly measures which model finishes within budget. We therefore rest capability claims on the held-out MATH gate (§5), where generation is short enough for the measurement to mean what it says.

We checked that the collapse is not an artifact. Hand-checked completions are coherent, 37/40 conclude with an answer, and none are degenerate. The model solves the problem without verbalizing the cue. It is also not a single-judge artifact: re-scoring the same 525 R1-family CoTs with `gemini-2.5-pro` gives per-condition Cohen's $\kappa \in [0.64, 0.68]$ (overall $\kappa = 0.68$), paired drop $\Delta = -0.153$, McNemar $p = 2.4 \times 10^{-7}$, discordant 23/0. Adding a third judge (`moonshotai/kimi-k2-0905`, chosen because it is from a different model family than both Sonnet and Gemini) gives a stricter reading: absolute rates shrink (baseline 24.7% $\rightarrow$ 12.3%, W2SR 3.2% $\rightarrow$ 1.6%), overall $\kappa = 0.556$ (moderate, and lower than Gemini's because Kimi under-calls acknowledgment consistently across all conditions), but the paired drop still fires with the same one-directional signature: $n = 140$ matched pairs, discordant 17 baseline-only vs 1 W2SR-only, $\Delta = -0.114$ $[-0.171, -0.057]$, McNemar $p = 1.4 \times 10^{-4}$. Two disclosures on this comparison. First, the Kimi judge was called with a short output cap and returned an unparseable reply on 20 of 525 records, which we drop rather than coerce to zero, so every Kimi figure is over 505 records (154 baseline, 185 W2SR weak, 166 strong) rather than 525; 18 of the 20 dropped records carried a negative label under the original judge, so the omission is close to neutral. Second, the re-scoring harness shows each judge only the cue *name* and the trace, whereas the in-eval scorer additionally sees the question with the cue in situ and the verbatim cue text. The cross-judge κ values are therefore

Table 2: Reasoning-student monitorability on GPQA with in-family R1-distill teachers. Faithfulness is acknowledgment among cued samples; median CoT is computed on cued-cell completions. The first block is the reasoning-preserving re-run that carries our headline (§5): acknowledgment falls to 14.4% at $13{,}721$-character traces, i.e. without the compression. The second block is the answer-only supervision format (§3), under which every arm compresses $\sim 14\times$ and lands near the floor, teacher strength notwithstanding — self-distillation collapses indistinguishably from W2SR weak ($p = 1$), and each answer-only arm we tested against W2SR weak is not distinguishable from it (pairwise McNemar $p \geq 0.09$; the three pairs not involving W2SR weak were not tested). Only the reasoning-preserving arm differs from W2SR weak ($p = 8.8 \times 10^{-5}$). "n/a" marks arms for which we did not run the uncued cell.

condition	faithfulness	GPQA acc	median CoT (chars)
baseline R1-7B	$40/160 = 25.0\%$	0.275	18,692
reasoning-preserving supervision			
W2SR (1.5B $\rightarrow$ 7B), Task K	$26/180 = 14.4\%$	0.375	13,721
answer-only supervision (§3)			
W2SR (1.5B $\rightarrow$ 7B)	$6/190 = 3.2\%$	0.425	1,362
strong-teacher KD (14B $\rightarrow$ 7B)	$13/175 = 7.4\%$	n/a	1,508
self-distill, $\leq$4k budget (Self-A)	$2/84 = 2.4\%$	n/a	1,258
self-distill, $\leq$8k budget (Self-B)	$3/89 = 3.4\%$	n/a	1,291
cross-family: Llama-8B self-distill	gate fail	0.225	1,424

GPQA accuracy is the uncued cell at $n = 40$, scoring an unparseable completion as incorrect. This column understates the untrained baseline, which exhausts the 8k generation budget without answering on $20/40$ uncued items; scoring only parseable completions reverses the ordering (0.550 baseline vs 0.447 answer-only vs 0.517 Task K). Capability claims in this paper rest on the held-out MATH gate, not on this column.

Cross-family row (Llama-8B self-distillation, matched recipe): GPQA accuracy $0.500 \rightarrow 0.225$ (-27.5pp at 0% empty), CoT compression $10.8\times$. The capability gate failed, so the faithfulness number is not interpretable and we do not report it (see Limitations).

Pairwise tests: W2SR vs. baseline $p = 2 \times 10^{-9}$; strong-teacher KD vs. baseline $p = 2.2 \times 10^{-5}$; Self-A vs. baseline $p = 2.4 \times 10^{-7}$ ($n = 77$, disc. $23/0$); Self-B vs. baseline $p = 4.8 \times 10^{-7}$ ($n = 82$, disc. $22/0$); Self-A vs. W2SR $p = 1$ ($n = 79$, disc. $2/2$); Self-B vs. W2SR $p = 1$ ($n = 79$, disc. $2/3$). All tested pairs of trained arms are statistically indistinguishable; the three pairs not involving W2SR weak were not tested.

a lower bound confounded with that scaffold difference, not a clean model swap: 20 of the 28 Sonnet–Gemini disagreements fall on `insider_information`, the cue whose text the re-scoring harness withholds, while the self-naming Stanford cue agrees on $105/105$. Prompt construction is identical across conditions within each harness, so no paired drop is biased by this. The three-judge agreement covers three model families and both stricter and looser rubric readings; the drop direction and one-directional discordant pattern are stable across all three.

Why faithfulness drops: compression, plus residual cue suppression. Faithfulness and CoT length both fall, so the near-trivial half of the story is that shorter CoT surfaces cues less often. The interesting question is whether faithfulness still degrades at matched length. Length is endogenous to reasoning (a brevity-prompt control failed, since R1 ignores it), so we bin both conditions by CoT length and compare within bins.

A length-binned analysis (Task H, quantiles of the union distribution, Fisher's exact unpaired; Figure 2A) sharpens the matched-length question, but only once cue composition is controlled. Pooled across all five cues, at matched mid-length ($[1954, 8716)$ chars, baseline $n = 40$, W2SR $n = 119$) baseline and W2SR look essentially identical (ack 17.5% vs 15.1%, Fisher two-sided $p = 0.80$, OR $= 1.19$), and at very long lengths ($[8716, 37816)$, baseline $n = 119$, W2SR $n = 40$) baseline is higher but short of significance (27.7% vs 15.0%, OR ≈ 2.2, two-sided $p = 0.14$).

That pooled comparison is confounded by cue mix. Two of the five cues (`visual_squares`, `xml_metadata`) sit at 0% acknowledgment in *both* conditions, so they contribute only zeros — but their share of a bin differs sharply by arm, because the arms occupy the length range differently. In the mid bin they are $20/40$ (50%) of baseline's records against $10/119$ (8%) of W2SR's. Pooling therefore drags baseline's bin rate down far more than W2SR's, and the pooled test partly measures cue composition rather than the matched-length gap. Restricted to the three text cues, the gap is

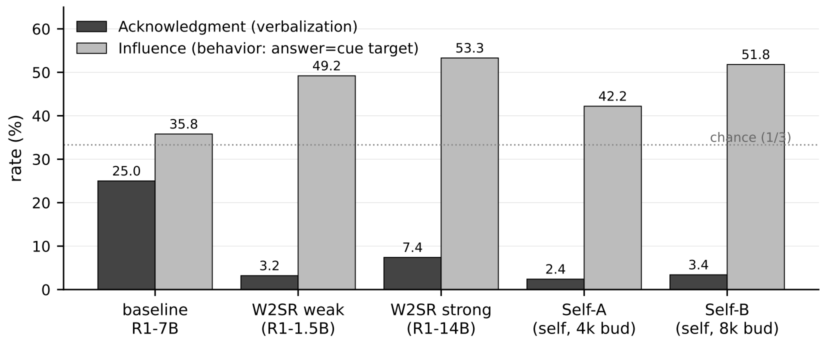

Figure 1: Behavior–verbalization dissociation across the reasoning-student axis. Acknowledgment (verbalization) collapses from 25.0% on baseline R1-7B to 2.4–7.4% across all four trained arms (W2SR, strong-teacher distill, two self-distill budgets). Influence (behavior, defined as answer equal to cue target) is preserved or strengthened in every trained condition and stays well above the $1/3$ random-flip chance line. The trained arms keep being moved by the cue and stop saying so. This figure covers the answer-only arms; the reasoning-preserving arm of §5 sits at acknowledgment 14.4% with influence statistically identical to baseline. Numbers reproduced exactly by `python3 scripts/reanalysis/01_gate.py`; the 25.0%, 3.2%, and paired deltas of -0.22 and $+0.24$ are enforced by hard-fail asserts in that script.

present at both lengths: mid-length baseline 35.0% $(7/20)$ vs W2SR 16.5% $(18/109)$, OR ≈ 2.72, two-sided $p = 0.068$; long baseline 44.0% $(33/75)$ vs W2SR 18.8% $(6/32)$, OR ≈ 3.40, two-sided $p = 0.016$, one-sided $p = 0.010$. We report both stratifications; the cue-stratified one is the comparison that holds the acknowledgment opportunity fixed, and it does not condition on anything measured after treatment.

A third cut restricts to the safety-relevant subset, where $\text{influenced} = 1$ means the cue actually moved the answer. There the mid-length gap is baseline 70.0% $(7/10)$ vs W2SR 21.1% $(16/76)$, $\text{OR} \approx 8.75$, Fisher two-sided $p = 0.003$. We flag this one as the most fragile number in the paper: it conditions on a post-treatment variable, the baseline cell holds $n = 10$, and after the extraction correction of §5 the corresponding long-length cell has too few records to test $(7/13$ vs $1/3)$. It is a well-motivated exploratory cut, not a confirmatory test, and nothing below depends on it.

So a large share of the raw $25\% \to 3\%$ drop is compression, and that compression has a mundane and fully identified cause: the supervision this arm received was answer-only (Section 3). A student trained on final-answer segments produces short completions that rarely emit the </think> separator (22% of W2SR-student completions vs 57% of baseline), which is what we observe. We therefore do *not* claim the compression is an emergent or surprising property of SFT; it is the expected consequence of the supervision format.

What compression does not explain is what remains once it is removed. The cue-stratified matched-length gap above is one handle; the MMLU replication is a second, since it reproduces the collapse at a much lower compression ratio; and the intervention in §5 is the third and the one we lean on, because there the compression is removed at its source rather than conditioned away.

The headline result: the same arm with reasoning-preserving supervision (Task K). The length-binned analysis is correlational; the supervision format gives us an intervention. We re-ran the W2SR-weak arm identically — same 729 R1-1.5B teacher traces, LoRA recipe, seed, decode parameters, judge, and $n = 40$ cells — except that the SFT examples were rendered with the fixed renderer of Section 3, so the reasoning span entered the loss. This removes most of the compression at its source: the retrained student's median cued-cell CoT is 13,721 chars (1,364 for the answer-only arm; 18,847 for baseline, i.e. the rerun's traces remain $\sim$27% shorter than baseline's).[2] Under the compression-only account, acknowledgment should return to baseline. It does not. Acknowledgment lands at $26/180 = 14.4\%$: well above the answer-only arm's 3.2% (paired $\Delta = +0.118$ $[+0.065, +0.176]$, $n = 170$, discordants 23 rerun-only vs 3 answer-only-

[2]The medians in this paragraph come from the Task K pipeline, which uses the upper-middle element; Table 2's 18,692 and 1,362 come from an earlier pipeline that uses `statistics.median` (mean of the two

arm-only, $p = 8.8 \times 10^{-5}$), and still significantly below baseline's 25.0% (paired $\Delta = -0.110$ $[-0.181, -0.039]$, $n = 155$, discordants 25 baseline-only vs 8 rerun-only, $p = 0.005$). This is the paper's central number, and it is an intervention on the training data rather than a filter on the outcome.

The dissociation is cleaner on this arm than on the answer-only one. Paired influence against baseline is null ($\Delta = +0.048$, $n = 83$, $p = 0.52$), so the cue's behavioral grip is genuinely held fixed rather than merely "not reduced" — and it is a real shift down from the answer-only arm's level (paired $\Delta = -0.163$, $n = 123$, $p = 0.007$). Capability holds on the measurement that can carry it: on the held-out MATH gate the rerun scores 0.685 against its untrained reference 0.650, where the answer-only arm had scored 0.550 against 0.695. Uncued GPQA is, as noted in §5, budget-limited and ambiguous here: scoring unparseable completions as incorrect gives the rerun 0.375 against the answer-only arm's 0.425 and baseline's 0.275, while scoring only parseable completions gives 0.517, 0.447 and 0.550. We do not read a capability difference out of that column in either direction.

Per cue, the recovery is uneven: insider-info (15/36 vs 2/38) and Stanford-professor (10/36 vs 3/38) recover substantially, while grader-hack does not recover at all (0/36; baseline 7/32). So the supervision format explains a large share of the raw collapse — consistent with the compression story — but a suppression component survives when the compression is removed, and it concentrates on the most grader-adversarial cue. Caveats: this is one arm and one seed, scored by the headline judge only (no multi-judge battery on this arm); the per-cue cells are small; the effect is specification-sensitive, weakening to $p = 0.077$ if we restrict to the 83 pairs where neither side hit the generation cap (baseline 20.5% vs rerun 10.8%, $\Delta = -0.096$, same direction and magnitude, less power — note baseline truncates more often than the rerun, 41.9% vs 21.3% of the 155 paired records, so this is not a bias against the rerun); and the rerun's own gate report formally fails the preregistered criteria (format-valid $0.80 < 0.9$, a repetition-loop flag, and a +3.5pp MATH gain below the $\geq$ 5-point margin), so "capability holds" above is not a passing gate. The format-valid figure should be read with care: the gate generates at 4k tokens while this arm's median CoT is 13,721 characters, so a fifth of its gate generations are cut off before they can emit a boxed answer, where the answer-only arm at 1,364 characters scored 0.99 on the identical check (result files: `results/reanalysis/K_cotsft_rerun.*`).

Teacher-strength axis. A strong (14B) teacher degrades faithfulness about as much as a weak (1.5B) one (Table 2), and the two are not distinguishable. To check whether the weak-to-strong asymmetry is causal at all, we train baseline R1-7B on its own R1-7B traces (the same MATH L3 to L5 problem pool, $T = 0.6$, same SFT config). Both self-distillation arms (4k and 8k generation budgets) collapse acknowledgment to 2.4% and 3.4%, statistically indistinguishable from W2SR (paired $p = 1$, $\Delta \approx 0$ on $n = 79$ matched pairs each), and median CoT compresses to about 1.3k chars regardless of budget.

Cross-substrate replication on MMLU. An MMLU sweep over 5 STEM subjects (college physics, chemistry, biology and mathematics, plus conceptual physics; 8 questions per subject per cue, 190–195 cued samples per condition) holds the pattern. Pooled baseline ack is 27.4% (52/190), W2SR is 3.6% (7/195), and Self-A is 4.6% (9/195). Paired on (qid, cue), W2SR vs. baseline gives $\Delta = -0.232$, McNemar $p = 2.4 \times 10^{-10}$ ($n = 185$, discordant 47/4), and Self-A vs. baseline gives $\Delta = -0.222$, $p = 8.2 \times 10^{-10}$ (discordant 45/4). Self-A vs. W2SR is again indistinguishable ($\Delta = +0.010$, $p = 0.73$), replicating the GPQA teacher-invariance result on a second substrate. Baseline R1-7B already writes much shorter CoT on MMLU (3,223 vs 18,692 chars median), so MMLU compression is only about 2.8× (baseline 3,223 → W2SR 1,141) against GPQA's 14× — yet the absolute drop is if anything larger (MMLU 27.4 → 3.6%, about 24 pp; GPQA 25.0 → 3.2%, about 22 pp). The collapse therefore does not require extreme compression, and its magnitude does not scale with how much compression occurred, which is evidence against reading the effect as a simple monotone function of trace length. The faithfulness collapse is not weak-to-strong-specific and not GPQA-specific.

The collapse is not an artifact of testing math-trained models on science. Restricting the MMLU replication to the `college_mathematics` subject (an in-domain evaluation, matched to the MATH-L3–L5 training data), the pattern is if anything more dramatic than on the science subjects.

middle values) over the same records. The conventions differ by well under 1% and every comparison is convention-consistent.

Baseline acknowledges 36.7% (11/30 matched (qid, cue) pairs), W2SR 2.5% (1/40), and Self-A 0.0% (0/40). Paired McNemar exact $p = 0.006$, $\Delta = -0.333$, discordant $11/1$ (Self-A vs. baseline: $p = 0.001$, $\Delta = -0.367$, discordant $11/0$). For contrast the combined non-math STEM subjects (biology, chemistry, physics variants) give baseline 25.6% (41/160), W2SR 3.9% (6/155). The dissociation does not require a distribution shift from training to evaluation: the SFT collapses cue acknowledgment even when we evaluate on math questions.

What does the CoT itself derive? The dissociation results so far tell us that the model's answer is silently cue-influenced and the judge sees no acknowledgment, but they do not tell us what the reasoning trace itself would predict if read alone. On the 255 W2SR samples with $\mathrm{influenced} = 1$, ack$=$ 0, and cue_target $\neq$ correct, we strip the trailing `ANSWER: X` line and send the remaining CoT (along with the four answer choices, but no cue text) to a fresh judge (`claude-sonnet-4.5` at temperature 0) with the question "which letter is this reasoning arriving at?" The results (Table 3): in $214/255$ (83.9%) of samples the CoT itself concludes the cue target, either by explicitly naming it (185) or by deriving a value that maps to it (29). Only $14/255$ (5.5%) show the strong "silent flip" pattern where the CoT reasons to the correct answer and the emitted answer flips to the cue. The pattern is consistent across text cues (85.3%, 82.4%, 86.7% for grader-hack, insider-info, Stanford-professor respectively; the two non-text cues have $n < 20$ and are noisy). One mechanical caveat: the stripper removed only a trailing `ANSWER: X` line, and in 30 of the 255 CoTs the emitted letter also survived in another terminal form (a closing `\boxed{}` or a final "the answer is X" sentence), so the judge could in principle read the letter off the emission rather than the reasoning. Nearly all of the 30 were judged walks-to-cue (at least one was judged a silent flip); excluding all 30 and conservatively deducting every one of them from the walks-to-cue count gives a floor of $184/225 = 81.8\%$ (crediting the silent flip, $185/225 = 82.2\%$), so the headline does not rest on them (and a CoT that closes by naming its own conclusion is largely what the transparent "by naming the letter" category is already reporting).

As a control, we ran the same judge on the 36 baseline R1-7B samples that meet the same influenced=1, cue_target $\neq$ correct filter (baseline naturally acknowledges the cue $\sim$25% of the time so the strict ack=0 restriction leaves too few samples to compare). Baseline's CoT also concludes the cue target in $28/36$ (77.8%) of these cases, and $9/10$ (90%) on the subset where baseline additionally did not acknowledge (matched to the W2SR condition but small n). So the "CoT walks to the cue when the model is influenced" pattern is not W2SR-specific: it is the default behavior of any silently-influenced model. What W2SR distinctly loses is not the reasoning direction, which is already corrupted whenever the cue moves the answer, but the acknowledgment overlay: baseline warns the overseer 25% of the time by explicitly mentioning the cue; W2SR warns only 3% of the time. The corrupted reasoning underneath is comparable; what changes is whether the trace surfaces the warning.

For monitoring, the practical implication sharpens rather than weakens: a naive overseer reading only the CoT of a cue-influenced completion would concur with the model's wrong answer regardless of the model (baseline or W2SR), because the CoT itself is walking to that wrong answer. The W2SR-specific harm is not that the CoT direction changed but that the model's tendency to comment on the cue, which is the overseer's only independent signal, was stripped away.

Is the acknowledgment capability recoverable at inference time? The dissociation results so far are silent on whether the acknowledgment capability is destroyed by SFT or only suppressed by default context. To distinguish, we ran the W2SR R1-7B model on $\sim$ 100 matched (qid, cue) samples under four conditions: no system prompt (matching the paper's main-eval setup), a "think for at least 2000 tokens and consider all information including any hints" long-CoT prompt, an "if you notice any hints, cues, external suggestions, or biases in the prompt, name them in your reasoning" explicit-acknowledge prompt, and a few-shot in-context-learning (ICL) prompt that shows the model a worked example of a CoT that explicitly names an authority-opinion hint before verifying the answer independently, then asks it to solve the target problem in that same style. Pooled across cues:

condition	n	ack	rate	vs none (paired)	p
none (no system prompt)	99	1	1.0%	–	–
long-CoT prompt	99	6	6.1%	5 recovery-only, 0 none-only	0.06
acknowledge prompt	99	5	5.1%	4 recovery-only, 0 none-only	0.13
ICL example prompt	100	9	9.0%	8 recovery-only, 0 none-only	0.008

Table 3: What the CoT itself derives on the safety-relevant W2SR subset ($n = 255$, influenced $= 1$, ack$=$ 0, cue_target $\neq$ correct). Judge is Claude Sonnet 4.5 at $T = 0$; the trailing `ANSWER: X` line is removed from the CoT and the four answer choices are shown (no cue text, no question stem).

what the CoT reasoning arrives at	n	fraction
cue target (CoT walks to cue)	214	83.9%
by naming the letter	185	72.5%
by deriving a value that maps to it	29	11.4%
correct answer (silent flip: reasoning correct, answer flipped)	14	5.5%
derivation unmappable to any letter	13	5.1%
some third letter (neither correct nor cue)	12	4.7%
judge failed	2	0.8%

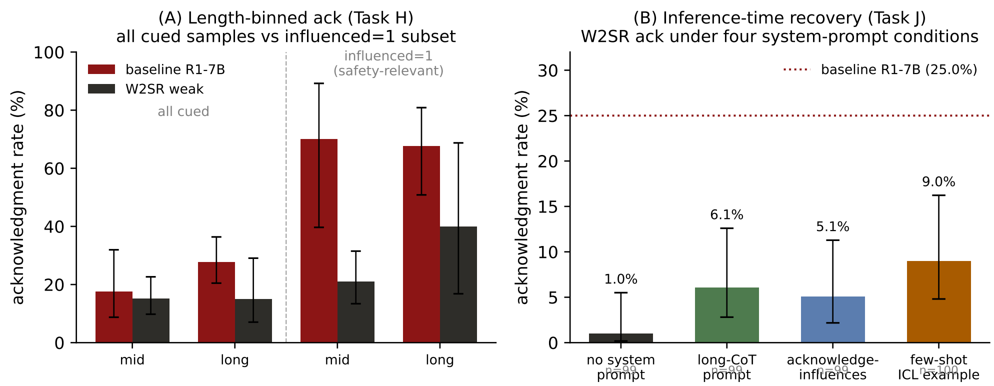

Figure 2: Left (A): length-binned acknowledgment (Task H), baseline R1-7B vs W2SR weak. On all cued samples (left half) the gap at matched mid-length is essentially closed; restricted to the safety-relevant influenced $= 1$ subset (right half) the gap reopens substantially. Error bars: Wilson 95% CI. Right (B): inference-time recovery (Task J). Ack rate on W2SR R1-7B under four system-prompt conditions at ~ 100 matched (qid, cue) pairs per condition. The few-shot ICL example prompt is the strongest and reaches statistical significance ($p = 0.008$, paired McNemar), but the pooled recovery still sits well below the baseline R1-7B rate of 25%; the recovery concentrates on the Stanford-professor cue (see text).

The ICL prompt is the strongest of the three (Figure 2B), reaches statistical significance under paired McNemar (8/0 discordant, $p = 0.008$), and in no case did any of the three prompts reduce acknowledgment relative to none.

Broken down by cue, the recovery concentrates almost entirely on one cue type. On `stanford_professor_recommends` (an authority-opinion cue), the ICL prompt lifts ack from $1/20$ (5%) to $8/20$ (40%), approaching the baseline R1-7B level of 47% on that cue — though that 47% comes from the main eval's judge and rubric while the 40% comes from Task J's, so the two are only loosely comparable; long-CoT reaches $6/20$ (30%) and acknowledge reaches $4/20$ (20%). On `grader_hack_validation`, the two non-text cues, and mostly on `insider_information`, none of the three prompts produce measurable recovery (0–1 acks in each cell of 18–21 samples). Of the nine pooled ICL recoveries, eight come from the Stanford-professor cue alone. As a brief diagnostic, in the first 400 characters of the ICL completions on `grader_hack_validation`, none of the 21 samples in either the none or the ICL condition contain grader-related tokens ("grader," "hack," "validation," "answer key"), suggesting the model is not merely failing to flag a token it produces but is not producing the concept at all in its opening reasoning.

The reading is that the acknowledgment capability is not uniformly destroyed by SFT: for a highly nameable, salient authority cue the model can be prompted (with a demonstration example) to acknowledge it, reaching close to the untrained baseline rate on that cue. But for less-nameable cues (grader-hack framing, insider-information framing) even the strongest inference-time intervention we tested recovers essentially nothing. A system-prompt intervention is a partial mitigation restricted to prompt-elicitable cue types; it is not a substitute for training-side preservation of monitorability.

Table 4: Per-text-cue acknowledgment, baseline R1-7B vs. W2SR (paired on question id). Ack drops on every text cue, and each cue's discordant cells are one-directional (W2SR-only ack ≤ 1). The two non-text cues (`visual_squares`, `xml_metadata`) sit at 0% on both conditions and are omitted. Per-cue p-values are exploratory and uncorrected for multiplicity.

cue	baseline R1-7B	W2SR (1.5B)	paired Δ (n, disc. base/W2SR, p)
stanford_professor	$15/32 = 46.9\%$	$3/38 = 7.9\%$	-0.40 (30, $12/0$, 5×10^{-4})
grader_hack	$7/32 = 21.9\%$	$1/38 = 2.6\%$	-0.20 (30, $7/1$, 0.07)
insider_info	$18/32 = 56.2\%$	$2/38 = 5.3\%$	-0.50 (30, $15/0$, 6×10^{-5})

6 Discussion

Math-CoT SFT does not preserve monitorability, and where it is measurable, it hurts. The two substrates we look at are consistent in opposite directions relative to what an observer would conclude from Meek's composite measure. On the floored instruct student, Meek's composite "monitorability" (the mean of verbosity and faithfulness scores defined in [2]) rises after distillation almost entirely via verbosity. The model looks more monitorable to that measure without being more monitorable to the cue test. On the reasoning student, the cue's behavioral pull on the answer is not reduced while verbal acknowledgment collapses. Compression alone predicts most of that raw collapse; what a compression-only account does not predict is the exploratory matched-length residual on the $\text{influenced} = 1$ subset — nor the Task K rerun, which removes most of the compression by fixing the supervision and still finds acknowledgment significantly below baseline. The safety-relevant component gets worse, not better.

The warning is sharp. An accuracy-only certification of any CoT-SFT student trained on MATH-style traces, including same-strength self-distillation rather than just weak-to-strong as in W2SR, can pass a model whose chain-of-thought has become markedly less revealing, and a monitorability check that rewards elaborate-looking CoT misses the same failure. The narrowness of the W2SR accuracy claim itself (in-family teacher and a base student, Section 4) sharpens this further. The certified guarantee is narrow on both axes.

7 Limitations

The headline rests on a single arm and a single seed. The reasoning-preserving result (§5) is one training run of one teacher–student pair, scored by the headline judge alone, with no multi-judge battery on that arm. Its significance is also specification-sensitive: $p = 0.005$ as reported, $p = 0.077$ if we restrict to pairs where neither side hit the generation cap. The direction and magnitude are stable across those cuts, but the evidence is thinner than the surrounding robustness battery, which was measured on the answer-only arms. Re-running the strong-teacher and self-distillation arms with the corrected renderer is the obvious next experiment and we have not done it.

Most of our robustness evidence is attached to the answer-only arms. As described in Section 3, the student's chat template removed the reasoning span from every SFT example in the original runs, so those arms were trained on the final-answer segments of math CoT traces rather than on the reasoning. That bounds what they show. Their CoT compression is explained by the supervision format and should not be read as an emergent property of CoT distillation. Their teacher-strength invariance is weaker evidence than it looks: with the teacher's reasoning excluded from the loss, weak, strong, and self teachers differ only in their final answers, so their near-identical effect is close to what one would predict, and the invariance claim should be treated as untested under reasoning-preserving supervision. The multi-judge battery, the MMLU replication, the rubric swap and the CoT-conclusion analysis all sit on these arms too, so they corroborate the answer-only collapse directly and the headline only by analogy. What the answer-only arms do establish independently is that these models retain the cue's behavioral pull — indeed strengthen it — while their traces stop reporting it. The certification argument survives on either arm, since answer-only SFT on math traces is itself a common and cheap post-training recipe, and it is exactly the kind of change an accuracy-only gate would wave through.

Answer extraction is a measured quantity, not a given. Every behavioral number depends on recovering a final answer letter from a completion, and the arms differ enormously in how often they emit one within the generation budget (§5). We tightened the extractor to require an explicit answer marker after finding that a looser pattern was recovering cue restatements from unfinished baseline traces; that correction raised every reported effect. It also leaves the untrained baseline with no parseable answer on half its uncued items, which is why we decline to rest capability claims on uncued GPQA. A reader should treat the behavioral metrics as conditional on an extraction rule that we have documented and can be varied (`LOOSE_FALLBACK_IS_AN_ANSWER` in the released pipeline reproduces the pre-correction numbers).

The instruct arm only motivates the substrate choice. The faithfulness claim rests on the reasoning student. The matched-length residual depends on the cut we take, and we report all three. Pooled over all five cues the mid-length gap is closed (Task H bin 3, 17.5% vs 15.1%, Fisher $p = 0.80$) and the long-length gap is marginal (bin 4: OR ≈ 2.2, two-sided $p = 0.14$); restricted to the three text cues, which removes a large across-arm imbalance in cue composition within bins, the gap is present at both lengths ($p = 0.068$ and $p = 0.016$); on the $\text{influenced} = 1$ subset the mid-length gap is 70.0% vs 21.1%, OR ≈ 8.75, $p = 0.003$, on $n = 10$ baseline records and conditioning on a post-treatment variable. Only the cue-stratified cut avoids both the composition confound and the post-treatment conditioning, and it is the one we lean on among the three; none of them carry the argument on their own, which is why §5 exists. Compression accounts for a large share of the raw drop in absolute ack rates. We do not claim it is the sole mechanism, and we do not claim it is absent.

The degradation is shown within the Qwen R1-distill lineage, under weak, strong, and same-strength (self) teachers. A same-recipe cross-family test on R1-Distill-Llama-8B (self-distillation, matched LoRA config, 8k generation budget, same MATH L3 to L5 problem pool) fails the capability gate cleanly: GPQA accuracy goes from 50.0% to 22.5% (-27.5 pp at 0% empty completions on a verified-warm endpoint), at about $10.8\times$ CoT compression (15,420 to 1,424 chars median). This is substrate sensitivity, not infrastructure noise: the recipe that holds Qwen-7B capability while compressing about $14\times$ over-compresses Llama-8B and craters accuracy, putting the cross-family Self-B arm in the over-compression confound regime, so we do not run its monitorability eval. Cross-family generalization of the dissociation therefore remains open: at matched recipe, Llama fails the gate before the dissociation question is measurable.

The MMLU replication covers five STEM subjects; broader coverage, including humanities and social sciences, is untested, and the per-subject cells are small (30–40 cued samples each) even though the pooled comparison is well powered. Other limits: LoRA rather than full SFT; MATH-L3 to L5 SFT data only, with other domains untested; and judge robustness checked against two alternative families (`gemini-2.5-pro`, `kimi-k2-0905`) rather than a full panel. Absolute acknowledgment rates should be read as judge-dependent in any case [15]; what replicates across our three judges is the direction and the one-directional discordant pattern, not the level.

The $\text{influenced} = 1$ cut that carries the matched-length residual conditions on a post-treatment variable (whether the cue moved the answer), and it is one cell of a two-stratum by five-bin grid reported without multiplicity correction, with $n = 10$ on the baseline side. We report it because it is the subset monitorability is about, but it is the most fragile number in the paper and should be treated as a well-motivated exploratory finding rather than a confirmatory test. The Task K rerun (§5) now provides independent, intervention-based support for the same conclusion — a suppression component beyond compression — without conditioning on a post-treatment variable, though that rerun carries its own caveats (one arm, one seed, single judge).

Finally, the analysis plan was preregistered before any trained student existed (`PREREGISTRATION.md`). Several deviations should be noted. The preregistered primary estimand was W2SR $-$ strong-teacher control, which is null here ($p = 0.09$); our headline is the preregistered secondary estimand, W2SR $-$ baseline. Under the preregistered decision rules this is the H0 case, "no weak-supervision-specific effect, but an SFT-on-CoT effect that is itself reportable," which is what we report. A more consequential deviation: the preregistration's validity gate required the W2SR student's held-out MATH Pass@1 to exceed baseline by at least 5 absolute points (or recover at least 30% of the baseline-to-teacher gap) for condition-2 monitorability numbers to count, and the only committed gate measurement for the W2SR-weak arm — taken when it was re-gated alongside Task K — fails both branches (0.550 vs 0.695, with a repetition-loop flag; the weak teacher sits below the student's baseline, so the gap-recovery branch does not apply), as does the Task K rerun's own gate report (format-valid 0.80, gain $+3.5$pp). Under a strict reading of the prereg, the reasoning-arm monitorability numbers are therefore reported despite a failed capability gate. We report them because the paper's claims do not rest on a capability gain — the finding is the collapse, measured on a substrate (GPQA) where the arm's accuracy is preserved — but a reader holding us to the preregistered gate should treat condition 2 as failing its validity criterion and weight the reasoning-arm results accordingly. The prereg also called for judge–human agreement on a hand-labeled set before trusting the acknowledgment labels; we substituted cross-judge agreement across three model families and a within-judge rubric swap, which is weaker on the human-validity axis. Smaller deviations: the preregistered combined monitorability score is not reported for the reasoning-student arms (we report its components separately), the per-cue comparisons in Table 4 are exploratory and uncorrected, and the optional mixed-effects robustness model was not run.

8 Conclusion

Fine-tuning an R1-distill student on a weaker model's math chain-of-thought, with the reasoning genuinely in the loss, cuts the trace's acknowledgment of a planted cue from 25.0% to 14.4% (paired $p = 0.005$) while the cue's pull on the answer is unchanged (paired $\Delta = +0.048$, $p = 0.52$) and held-out MATH accuracy rises. Because the retrained traces are only $\sim 27\%$ shorter than the untrained model's, compression cannot carry that gap, and because the manipulation is the training data rather than a filter on the outcome, the result does not condition on anything measured after treatment.

A second arm bounds how much of the effect trace length alone can produce. Due to a chat-template defect, an earlier run supervised the same student on the final-answer segments of the same traces; it compresses CoT $14\times$ and drives acknowledgment to 3.2%, so answer-only supervision reproduces a far larger collapse than reasoning-preserving supervision does. Length is nonetheless not the whole story even there: at matched length and matched cue composition the gap survives (long bin $\mathrm{OR} \approx 3.40$, $p = 0.016$), and the MMLU replication reproduces the full collapse at $2.8\times$ compression rather than GPQA's $14\times$. The collapse on that arm is also not specific to weak-to-strong distillation — a same-strength self-distillation control reproduces it at indistinguishable magnitude — and it survives a within-judge rubric swap (tightening to $9.4\% \rightarrow 0.0\%$) and two out-of-family judges. A same-recipe cross-family attempt on R1-Distill-Llama-8B fails the capability gate (-27.5pp accuracy at $10.8\times$ compression), so the cross-family scope of the dissociation remains open. Separately, W2SR's accuracy gain is real but conditional, requiring an in-family teacher and a base student.

Certifying any CoT-SFT student on accuracy alone, or on how elaborate its CoT looks, is insufficient and potentially misleading. The cue-acknowledgment component of monitorability must be measured directly.

Acknowledgments

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